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In *vitro* sterilization protocol for micro-propagation of *Withania somnifera* L. Dunal

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Abstract : For obtaining contamination free cultures the most important step is sterilization. In the present study the sterilization protocol was standardize for *Withania somnifera*. Effect of mercuric chloride in different concentration for different time period was checked for shoot tip explant. . 0.025%, 0.05%, 0.1% and 0.2 of $Hgcl_2$ for 5, 10, 15 minutes for shoot tips were checked. Its found that high concentration with long time period was harmful for explant causing large rate of deadening .10% $Hgcl_2$ for 10 mins for shoot tips found best sterilization chemical after 15 mins thorough wash with 0.1% bavistin and at last washed with double distilled water. The results revealed a significant relationship between the concentration and time period of $Hgcl_2$ and surviving % of explant of Ashwagandha in tissue culture.

Key Words: Sterilant, Contamination, Surface sterilization,Mercuric Chloride.

INTRODUCTION

Withania is an important member of family Solanaceae. Twenty three species of the genus of *Withania* have been reported (Negi et. al., 2006). *Withania somnifera* is commercially important because of its polypharmaceutical potential. Plant has been used medicinally in treatment of tuberculosis, rheumatism, inflammatory conditions, cardiac diseases, and it is used as a general tonic, anti-stress drug and as anti-tumor, anti-biotic, anti-convulsant, Central nervous system (CNS) depressant agent. Biotechnological tools are important for multiplication and genetic enhancement of medicinal plants by adopting techniques such as selection, multiplying, *in vitro* regeneration, genetic transformation and conserve the critical genotypes of medicinal plants (Tripathi and Tripathi, 2003; Rout et. al., 2000).

Sterilization is the process of making the explants contamination free before establishment of cultures. The first condition for success of a cultures is aseptic. To maintain an aseptic environment, all culture vessels, media and instruments must be sterilized. Besides these sterilization of explant is one of the most important step in *in vitro* technique. Several sterilization agents are used to decontaminate the explants in tissue culture but these sterilants are also toxic to plant tissues, hence proper concentration of sterilant and duration of exposing the explants has to be standardized to minimize explant injury and achieve better survival.

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MATERIALAND METHODS

Shoot tips of *Withania somnifera* from *in vivo* grown plants were used as explant for the present study. Shoot tips were properly washed with tap water for removal of sand, soil etc. Then the explants were kept in running tap water for 10 mins. 0.1% Bavistin was used as fungicide for 15 mins. After that the explants were washed properly with distilled water for 3 times. Rest of the treatments were given in Laminar airflow. Mercuric chloride ($Hgcl_2$) was used as main sterilant for the present study. Different concentration of $Hgcl_2$ were prepared in .025%, .05%, .1% and .2% with autoclaved distilled water. With all the 4 concentration of $Hgcl_2$ the explants were surface sterilized with three selected timings of 8, 10, 12 minutes with constant and thorough shaking.

Table 1.These are the treatments used for this work.

T_1	–	0.025%	–	5 minutes
T_2	–	0.05%	–	"
T_3	–	0.1%	–	"
T_4	–	0.2%	–	"
T_5	–	0.025%	–	10 minutes
T_6	–	0.05%	–	"
T_7	–	0.1%	–	"
T_8	–	0.2%	–	"
T_9	–	0.025%	–	15 minutes
T_{10}	–	0.05%	–	"
T_{11}	–	0.1%	–	"
T_{12}	–	0.2%	–	"

RESULT AND DISCUSSION

The present study was conducted to standardized the sterilization procedure of explants of *Withania somnifera*. Twelve different treatments were checked for the study.

Effect of infected cultures:

Result showed that with incensement of time period the infection rate is decreased in each concentration of $Hgcl_2$. The infection was notably much lower 04.16s% in 0.2% $Hgcl_2$ for 15 minutes (T_{12}). Low concentration 0.025% of $Hgcl_2$ for small time period 5 mins showed highest 77.08% infection rate (T_1).

Effect of sterile culture:

The data indicates that increasing the concentration and time period of $Hgcl_2$ the percentage of sterility was also increased and reached to 95.84% in T_{12} treatment.

Effect on viable cultures:

On increasing the concentration and duration of $Hgcl_2$ exposure it found that in T_{12} treatment rate of infection is lowest and rate of sterility is highest, while on the other hand rate of viability is lowest reaching 16.67%. Viability is found highest in 0.1% $Hgcl_2$ for 10 minutes (T_7) treatment of 85.41%.

Table: Effect of $HgCl_2$

Treatment	% of infection	% of sterility	% of viability
T_1	**77.08**	**22.92**	**20.83**
T_2	**58.33**	**41.67**	**39.58**
T_3	**45.83**	**54.17**	**52.08**
T_4	**27.08**	**72.91**	**43.75**
T_5	**52.08**	**47.92**	**47.91**
T_6	**33.33**	**66.67**	**62.50**
T_7	**10.41**	**89.59**	**85.41**
T_8	**08.33**	**91.67**	**37.50**
T_9	**35.41**	**64.59**	**58.33**
T_{10}	**20.83**	**79.17**	**54.16**
T_{11}	**06.25**	**93.75**	**45.83**
T_{12}	**04.16**	**95.84**	**16.67**

Suitable sterilization:

Thus from the present study it revealed that 0.1% $Hgcl_2$ for 10 minutes treatment followed by washed with double distilled autoclaved water for 3 times is suitable for shoot tips sterilization of *Withania somnifera* I. Sivanesan 2007, used 0.2% $Hgcl_2$ for 5 minutes after treatment of 10 mins Teepol solution and 1 mins Ethanol treatment.

CONCLUSION

From the results, it can be concluded that for *in vitro* culture of *Withania somnifera* 0.1% mercuric chloride for 10 minutes constant shaking is best.

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